

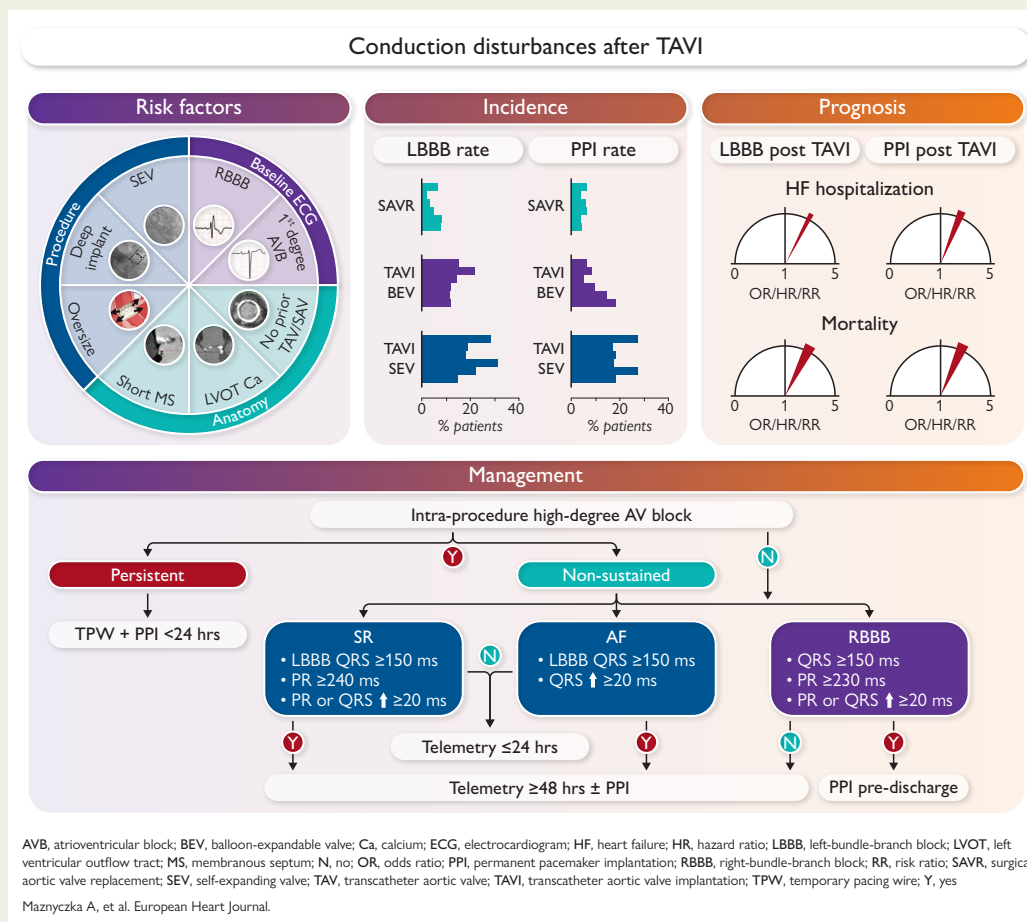
Conduction disturbances after transcatheter aortic valve implantation

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Graphical Abstract



Conduction disturbances after transcatheter aortic valve implantation: the risks factors, incidence, prognosis, and management.

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Abstract

Conduction disturbances and permanent pacemaker implantation remain the most common complications after transcatheter aortic valve implantation. The strongest predictors of conduction abnormalities and subsequent permanent pacemaker implantation after transcatheter aortic valve implantation include pre-existing right bundle branch block, a short membranous interventricular septum, deep transcatheter heart valve implantation, and valve type. Importantly, both new permanent pacemaker implantation and new left bundle branch block after transcatheter aortic valve implantation are associated with increased mortality and heart failure hospitalizations. As transcatheter aortic valve indications expand to lower risk and younger populations, with longer life expectancy, strategies to minimize the risk of conduction disturbances and optimize their detection and management become increasingly crucial. Refined transcatheter heart valve implantation techniques may be associated with a reduction in rhythm disturbances after transcatheter aortic valve implantation and anti-inflammatory treatments are under investigation. Ongoing trials are investigating the impact of beta-blocker withdrawal to prevent conduction abnormalities, electrophysiology studies for risk stratification, and conduction system pacing to prevent adverse cardiac remodelling. This review aims to provide an overview of the incidence, pathophysiology, and consequences of conduction disturbances after transcatheter aortic valve implantation, discuss preventive strategies, highlight the relevant ongoing studies, and provide an evidence-based framework for the management of this important clinical issue.

Keywords Conduction disturbance • Left bundle branch block • Permanent pacemaker • Right bundle branch block • Transcatheter aortic valve implantation

Introduction

Transcatheter aortic valve implantation (TAVI) is an established treatment for severe aortic stenosis, and with the expansion to younger populations, consequent permanent pacemaker implantation (PPI) gains prominence for lifetime management considerations.^{1–8} Despite new transcatheter heart valve (THV) iterations, the evolution of TAVI techniques such as high THV implantation, and the expansion to younger patients, conduction abnormalities remain common. After TAVI, the incidence of new permanent PPI and new-onset left bundle branch block (LBBB) varies widely between patient risk profiles, THV types, and healthcare settings. Importantly, both new PPI and new-onset LBBB post-TAVI are associated with increased mortality and heart failure hospitalizations.^{9–11} As TAVI indications expand to younger and lower-risk populations, with longer life expectancy, strategies to minimize the risk of conduction disturbances and optimize their detection and management become increasingly crucial. This review aims to provide an overview of the preventive strategies, incidence, pathophysiology, and consequences of conduction disturbances after TAVI, highlight ongoing studies, and provide an evidence-based framework for the management of this important clinical issue (*Graphical Abstract*).

Proximity of the aortic valve to the conduction system and anatomical variations

The His bundle penetrates the lower border of the membranous septum and runs superficially along the left ventricular (LV) side of the ventricular septum, giving rise to the left bundle branch, which is in close proximity to the aortic annulus (*Figure 1*). The close proximity of the aortic valve to the conduction system explains why conduction disturbances are common after TAVI. Of

note, inter-individual variability in the course and depth of the His and left bundle in the membranous septum can explain variation in PPI risk.¹² In a minority of people, the right bundle originates from the left side of the septum before traversing to the right septum, which explains the finding of new-onset right bundle branch block (RBBB) after TAVI.¹² The fibres compressed by a THV at the His bundle are destined to form either left or right bundle, due to longitudinal dissociation of these fibres; hence, the bundle with greater slowing relative to the other will determine the electrocardiogram (ECG) appearance.¹³

The THV may impinge directly on the conduction system, leading to mechanical insult that can progress with additional mechanical pressure [self-expanding valve (SEV)]. In addition, oedema resulting from the local inflammatory response, local ischaemia, and intramural haematoma may compress the conduction system, leading to transient, permanent, or delayed conduction disturbances.^{14,15}

Incidence of conduction abnormalities after transcatheter aortic valve implantation

The 30-day incidence of new PPI varies between studies, but remains consistently higher after TAVI compared with surgical aortic valve replacement (SAVR) (median 14% vs 5% in randomized trials)^{1–8,16–27} (*Figure 2*). The 30-day incidence of new-onset LBBB was reported in only four of these randomized trials [18% for CoreValve (Medtronic),²⁶ 15% for SAPIEN-XT (Edwards Lifesciences),⁴¹ 22% for SAPIEN-3,³ 12% for Octacore Myval (Meril Life Sciences),²³ and 8% for SAVR³]. Overall, new-onset persistent LBBB post-TAVI occurs in about 20% of patients,^{3,23,26,28–42} whereas after SAVR, the incidence is about 5%^{3,36–40} (*Figure 2*). New-onset RBBB after TAVI is rare, with an incidence of 0.3%–0.8%.^{13,43,44}

Of note, the PPI rates reported in studies may not solely reflect high-degree atrioventricular (AV) block, because some patients will have received PPI for other reasons, e.g. sick sinus syndrome or prophylactically in higher-risk populations. There is a lack of uniformity in prior trials and registries on the type of new conduction disorders that are reported as events. To facilitate comparisons between future studies, the Valve Academic Research Consortium (VARC)-3 states that PPI rates post-TAVI or SAVR should exclude patients with pre-existing pacemakers.⁴⁵ VARC-3 further proposes the following criteria to categorize the duration of LBBB after TAVI/SAVR: transient (resolved before discharge or <7 days), persistent (present at hospital discharge, or 7–30 days), or permanent (present >30 days after the index procedure).⁴⁵ Furthermore, VARC-3 categorizes the timing of occurrence of conduction disturbances after TAVI/SAVR as procedural (≤ 24 h) or delayed (> 24 h).⁴⁵

Predictors of conduction disturbances after transcatheter aortic valve implantation

Baseline ECG, anatomical, and procedural predictors of conduction disturbances after TAVI are summarized in [Figure 3](#).

Patient characteristics

Pre-existing RBBB, isolated or combined with left hemiblock, is present in about 10% of TAVI patients⁴⁶ and is a strong predictor of PPI (odds ratios 2.48–12.62 in observational studies).^{47–51} Pre-existing first-degree heart block^{49,52} or left anterior hemiblock^{50,52} has also been associated with PPI. There are scarce and discrepant data on pre-existing LBBB and PPI requirement post-TAVI; however, an association has been reported.^{53,54} Comorbidities that have been inconsistently associated with new PPI after TAVI include a history of atrial fibrillation,^{47,55} older age,^{46,48,55,56} LV ejection fraction (LVEF) < 50%,⁵⁷ and diabetes.⁵⁵

Anatomical characteristics

Calcification of the LV outflow tract (LVOT) has been associated with an increased risk of PPI post-TAVI^{58–60}; however, this observation is inconsistent.⁶¹ The pattern of calcium in the LVOT may be relevant, as some studies suggest heavy calcification below the left coronary cusp (LCC) and a low calcium burden below the non-coronary cusp (NCC) may create an imbalance in mechanical stresses exerted by the THV frame towards the NCC, resulting in more stress exerted near the conduction system.^{59,60} In addition, an eccentric³¹ or tapered LVOT⁶² and a large aortic annulus size⁵⁸ have been associated with PPI in some studies.

A short membranous interventricular septum (surrogate anatomical landmark for the aortic annulus to His bundle distance^{63,64}) is generally associated with higher PPI rates after TAVI^{63–66} and new-onset LBBB.^{31,67} A membranous septum length > 7 mm has a low PPI rate post-TAVI (<10%); conversely, a membranous septum length ≤ 3 mm carries a high risk of PPI (>20%).⁶⁵ On the other hand, a pre-existing SAVR or TAVI is associated with a lower risk of PPI, likely due to a protective effect of the bioprosthesis on the conduction system from the external forces of the new THV.^{56,68,69}

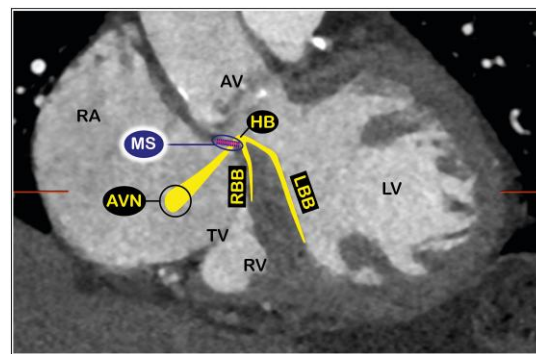


Figure 1 Schematic of the conduction system and the corresponding computed tomography landmarks. Contrast enhanced computed tomography coronal view showing the His bundle emerging at the inferior margin of the membranous septum, where it bifurcates exposing the LBB at the surface of the left ventricle, causing it to be susceptible to interference from the inflow of an implanted transcatheter aortic valve. The atrioventricular node is located in the right atrium, within the triangle of Koch, which is demarcated by the attachment of the septal leaflet of the tricuspid valve, the orifice of the coronary sinus, and the tendon of Todaro. The membranous interventricular septum is at the apex of the triangle of Koch on the right side, whereas on the left side, the membranous interventricular septum is just below the commissure between the non and right coronary cusps. The atrioventricular node continues as the bundle of His, which penetrates the lower border of the membranous septum, thereby entering the left ventricular surface in the left ventricular outflow tract. AV, aortic valve; LBB, left bundle branch; RBB, right bundle branch; RV, right ventricle

Procedural characteristics

Deep THV implantation is associated with new-onset LBBB^{31,70} and PPI.^{59,71–73} Intra-procedural AV conduction block is another predictor of PPI.⁷⁴ In particular, when the intra-procedural conduction disturbance occurs before THV implantation, the rate of persistence at 1 month may be higher.⁷⁵ Other procedural predictors of PPI include non-coaxial THV deployment,⁷⁶ larger TAVI size,⁶³ and TAVI type.^{24,48,58,77,78} Self-expanding valves are associated with higher PPI rates compared with balloon-expandable valves (BEVs)^{48,58} (median 19% vs 9% in randomized trials)^{1–4,7,8,16,19–27} ([Figure 2](#)). There is, however, a substantial variation between different SEV and BEV designs. In the COMPARE-TAVI-1 (SAPIEN-3 vs Myval THVs for TAVI) trial, the Octacore Myval BEV had a two-fold higher 30-day PPI rate compared with the SAPIEN-3 BEV (19% vs 10%, $P = .0002$),²⁴ possibly due to a different type and thickness of external skirt with the former. Furthermore, the JenaValve Trilogy system (JenaValve Technology) for aortic regurgitation has been associated with high PPI rates (19.6%–24.1%),^{77,78} possibly due to patients with pure aortic regurgitation having larger annular diameters and minimal leaflet calcification, which restricts THV expansion.

Oversizing with BEVs has been associated with a higher risk of PPI and controversial results have been reported with

SEVs.^{52,79,80} Findings from studies on the effect of balloon dilatation on PPI rates after TAVI are inconsistent.^{66,81–83} A meta-analysis of observational studies, which included a total of 5740 patients, found that with increasing use of post-dilatation, the effect of shorter membranous septum length on PPI rates increased; however, the effect was small.⁶⁶ Other meta-analyses of observational studies have reported discrepant findings on the effects of pre-dilatation on PPI rates,^{82,83} and two randomized trials comparing pre-dilatation with direct TAVI found no difference in PPI rates between the two groups.^{84,85} The increase in PPI rates observed in some studies might not be fully explained by pre- or post-dilatation, as confounders can affect these associations. Indeed, pre- and post-dilatation is often performed with severely calcified valves, in order to cross the valve and to achieve optimal THV expansion, and heavy LVOT calcification is itself associated with higher PPI rates post-TAVI.^{58,86} Importantly, pre- and post-dilatation with appropriately sized balloons is used as a strategy to improve THV expansion.^{87–89} Transcatheter heart valve under-expansion has been associated with adverse effects, including paravalvular leak, leaflet pin-wheeling, hypoattenuating leaflet thickening, late stroke, and mortality.^{90,91} Therefore, in the absence of robust evidence of balloon dilatation being associated with PPI, the appropriate use of pre- and post-dilatation should not be restricted.

Clinical implications of conduction disturbances after transcatheter aortic valve implantation

In the shorter term, conduction disorders after TAVI are associated with longer hospitalization, increased hospital costs,⁹² and peri-procedural complications related to temporary pacing wire (TPW) implantation or PPI, including immobilization, thromboembolism, bleeding, ventricular perforation, lead dislodgement, and infection. New-onset persistent LBBB after TAVI is not a benign outcome, as it has been associated with increased mortality at 1 year, heart failure hospitalizations, future complete heart block, and PPI (Figure 4).^{11,93,94} The adverse long-term prognosis associated with new-onset LBBB may be related to interventricular dyssynchrony and abnormal septal motion, reducing the LVEF, preventing LVEF recovery after TAVI and negative ventricular remodelling leading to heart failure.⁹⁸

Similarly, new PPI after TAVI is associated with increased mortality and heart failure hospitalizations at long-term follow-up (Figure 4).^{9,10,95–97} A higher percentage of right ventricular pacing ($\geq 20\%$ – 40%) is considered to predict worse prognosis.⁹⁹ The adverse long-term prognosis after PPI is likely due to electromechanical ventricular dyssynchrony, resulting from right ventricular apical pacing causing an abnormal LV activation sequence, leading to decreased cardiac output, higher residual LV volume, increased mitral regurgitation, negative LV remodelling, and reduced LVEF.⁹⁹ In addition, PPI post-TAVI may be associated with progression of tricuspid regurgitation, possibly due to tricuspid annular deformation, resulting in malcoaptation of the tricuspid valve leaflets, which may contribute to heart failure symptoms.¹⁰⁰ Of note, patients with new PPI post-TAVI

may not necessarily develop LV dyssynchrony, as many of these patients have no or low pacing burden at follow-up.¹⁰¹ There is a need for strategies to reduce conduction disorders post-TAVI and for careful evaluation of patients who should have PPI. Avoiding unnecessary PPI and optimizing programming to minimize right ventricular pacing or adopting a more physiological pacing strategy (conduction system pacing) is especially important in younger patients, where the long-term complications of PPI are likely to be higher.

Transcatheter aortic valve implantation procedure optimization to minimize conduction disturbances

Implant depth, coaxiality, and precise transcatheter heart valve positioning

Transcatheter heart valve implant depth^{63,102–104} and coaxiality⁷⁶ of the THV inflow relative to the aortic annulus are important determinants of new-onset conduction disorders and PPI after TAVI. A deeper implant^{102–104} and implant depth greater than the membranous interventricular septum length⁶³ predict an increased risk of PPI. Measurement of the membranous septum length pre-TAVI can help plan the optimal THV implant depth to avoid PPI (Figure 5). However, the membranous septum depth varies within the same patient, such that it is longer adjacent to the NCC than the right coronary cusp (RCC).¹⁰⁵ Membranous septum area has been shown to add incremental prognostic value for predicting conduction abnormalities post-TAVI over and above the membranous septum length.¹⁰⁵ Notably, the MIDAS (minimizing depth according to the membranous septum) approach, aiming for a pre-release depth with Evolut THVs < membranous septum length was shown to reduce the new PPI rate from 9.7% to 3.0% ($P = .035$) and the new-onset LBBB rate from 25.8% to 9.0% ($P < .001$).⁶³

Implanting the THV at a precise depth is facilitated by (i) computed tomography pre-TAVI for optimal fluoroscopic view selection for cusp visualization, (ii) accurate pigtail catheter position at the cusp nadir, (iii) aortography during valve deployment, (iv) rapid pacing to achieve adequate blood pressure suppression for stable THV release, (v) recognizing infolds in SEVs, and (vi) wire choice/manipulation to achieve coaxial THV deployment relative to the annulus. Of note, a higher THV implantation may complicate future coronary access in patients with low coronaries and shallow sinuses. Moreover, a high THV implantation creates a higher neo-skirt in future TAV-in-TAV implantations, due to upward displacement of the index TAVI leaflets by the new valve frame. Very high THV deployments (such as 0-mm depth) can predispose to valve embolization ('pop-out').

Radiolucent line technique for SAPIEN-3 deployment

For the SAPIEN-3 BEV, a high implantation technique uses a modified two-cusp view [right anterior oblique (RAO) caudal], modified to remove parallax in the THV frame, to visualize the radiolucent line.^{71–73} The radiolucent line, which is located at the superior aspect of the lowest set of stent struts of the crimped SAPIEN-3 valve, is then positioned at the nadir of the NCC during

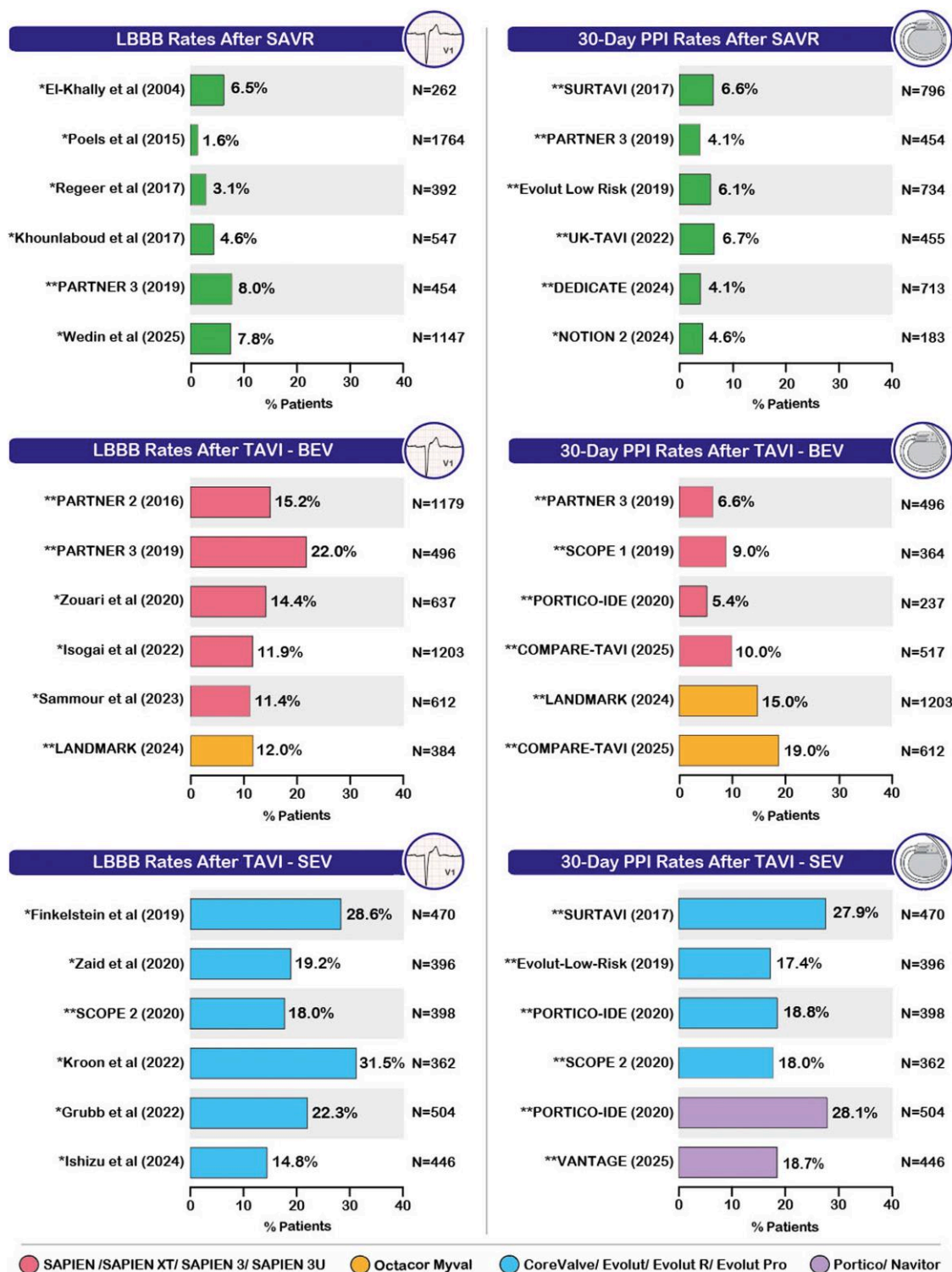


Figure 2 Rates of permanent pacemaker implantation and persistent left bundle branch block after transcatheter aortic valve implantation and surgical aortic valve replacement, in order of publication date, and according to valve type. *Observational study/registry.²⁸⁻⁴⁰ **Randomized controlled trial.^{3-6,8,16,17,20,22-24,26,27}

BEV, balloon-expandable valve; COMPARE-TAVI, SAPIEN-3 vs Myval transcatheter heart valves for TAVI; DEDICATE, randomized trial of TAVI vs SAVR in patients with severe aortic stenosis and low to intermediate risk of mortality; LANDMARK, randomized controlled trial of Myval transcatheter heart valve; LBBB, left bundle branch block; NOTION, Nordic aortic valve intervention; PARTNER, the placement of aortic transcatheter valves; PORTICO-IDE, Portico investigational device exemption; SAPIEN-3 U, SAPIEN-3 Ultra; SAVR, surgical aortic valve replacement; SEV, self-expanding valve; SCOPE, safety and efficacy comparison of two TAVI systems; SURTAVI, surgical replacement and TAVI; TAVI, transcatheter aortic valve implantation; VANTAGE, evaluation of transcatheter aortic valve replacement using the Navitor valve in a global investigation.

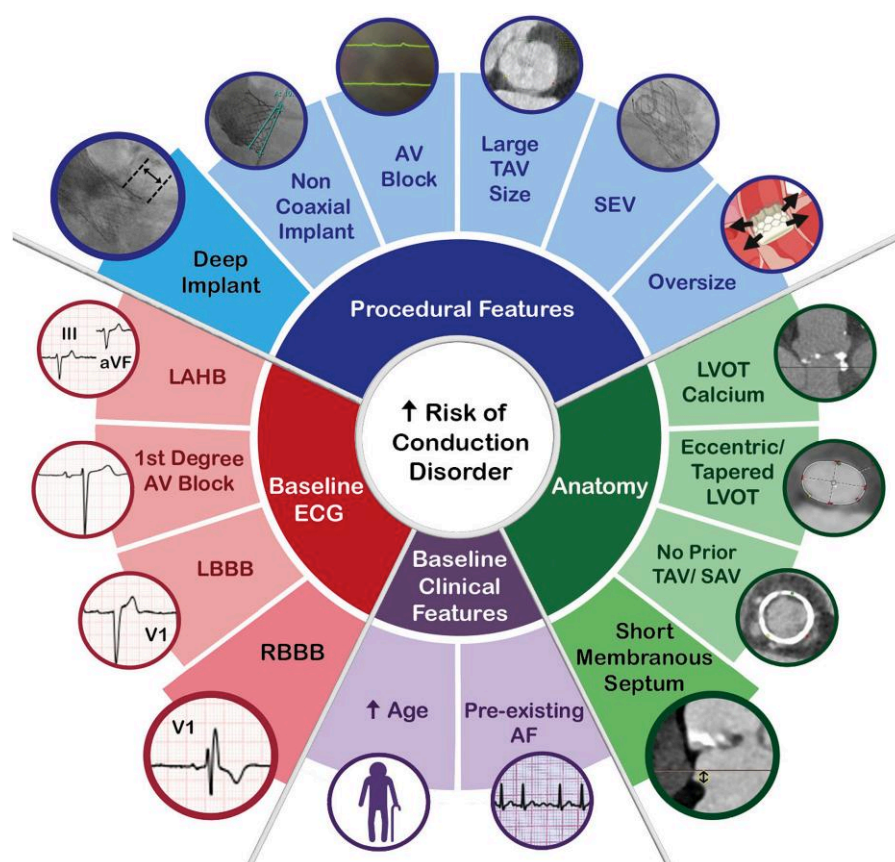


Figure 3 Predictors of conduction disturbance after transcatheter aortic valve implantation. AF, atrial fibrillation; AV, atrioventricular; ECG, electrocardiogram; LAHB, left anterior hemiblock; LBBB, left bundle branch block; LVOT, left ventricular outflow tract; RBBB, right bundle branch block; SAV, surgical aortic valve; SEV, self-expanding valve; TAV, transcatheter aortic valve

deployment⁷² (Figure 6). This technique has been shown to result in a higher THV position and lower incidence of new conduction disturbances after TAVI, compared with using the central balloon marker to guide implantation in the three-cusp view (mean implant depth: 1.5 vs 3.2 mm, $P < .001$ ⁷²; new-onset LBBB: 1.1% vs 7.0%, $P = .047$ ⁷¹; and new PPI: 2.9% vs 13.5%, $P < .001$ ⁷³).

Optimized implantation technique for self-expanding valves

For deploying the Evolut self-expanding THV, the cusp-overlap technique¹⁰⁶ exhibited relatively low PPI rates in the Optimize PRO study²⁹ and in a real-world registry of 54 014 patients⁵⁵ (9.2% and 10.8%, respectively, at 30 days post-TAVI). In the cusp-overlap technique, the RCC and LCC are overlapped (usually in a RAO view) to isolate the NCC. Parallax is eliminated from the THV by moving the fluoroscopic angle more caudal (or seldom cranial) (Figure 7). The advantage of starting SEV deployment in the NCC isolation view (i.e. RCC-LCC overlap view) is that the native aortic annulus and THV tend to be visualized in the same plane. The RAO view removes foreshortening from the LVOT, allowing more accurate assessment of the depth of the inflow of the THV frame relative to the nadir of the NCC. The targeted implant depth for the Evolut THV is 3 mm at the NCC (the level of the Evolut valve marker dot). When the

Evolut THV is 80% deployed, with full annular contact, the fluoroscopic angle is moved to a left anterior oblique (LAO) view to isolate the LCC, with the RCC and NCC overlapped. Parallax is removed from the THV, such that the three-valve marker dots are in line. This allows for depth assessment at the level of the LCC, aiming for a depth between 1 and 5 mm. Of note, the valve delivery wire tends to favour the inner curve of the ascending aorta with the Evolut THV; hence, simultaneous pushing on both the wire and the valve delivery system can centralize the wire position to avoid a non-coaxial excessively deep THV position on the side of the LCC. The cusp-overlap technique has also been shown to lower PPI rates with the Portico/Navitor SEV (Abbott Structural Heart) [from 18% (using the LAO view) to 12.6%].¹⁰⁸

Medical treatment to minimize conduction disturbances

Endothelial damage from THV implantation may result in leukocyte mobilization towards the injured area, inflammatory mediator release, increased vascular permeability, and local oedema, leading to conduction abnormalities. It is postulated that modulation of inflammation, e.g. with peri-procedural glucocorticoids, may reduce AV block post-TAVI (Table 1).

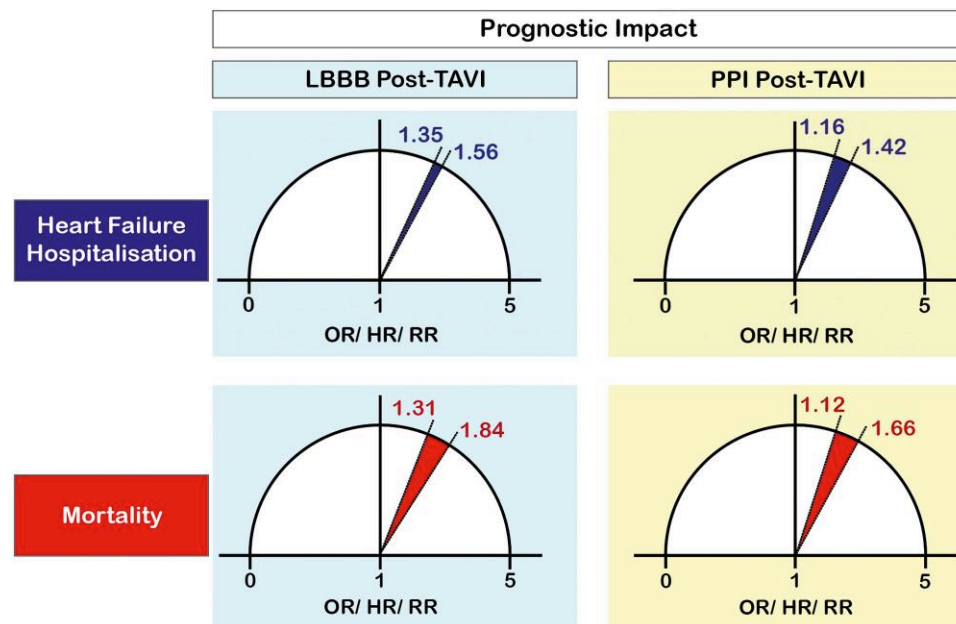


Figure 4 Range of hazard ratios, odds ratios, and risk ratios for associations with post-transcatheter aortic valve implantation mortality and heart failure hospitalizations at 1 year, from meta-analyses and large registries.^{9-11,93-97} LBBB, left bundle branch block; TAVI, transcatheter aortic valve implantation

Observational studies have not shown a consistent association between systemic corticosteroid exposure and AV conduction disturbances after TAVI.^{109,110}

In the randomized GLUCOTAVI (glucocorticoids in patients undergoing TAVI to prevent pacemaker implantation) trial, which included 100 patients, methylprednisolone 1 h before TAVI followed by prednisone for 5 days did not reduce the rate of PPI at 1 year.¹¹¹

In the Co-STAR (colchicine for patients with aortic stenosis undergoing TAVI) trial, peri-TAVI colchicine was found to lower the composite rate of new PPI or new-onset atrial fibrillation at 30-days compared with placebo.¹⁴ In contrast, a retrospective study found no difference in high-degree AV block or PPI rates in TAVI-patients with or without colchicine; however, the incidence of LBBB at 1 month was lower with pre-procedural colchicine (difference not sustained at 6 months).¹⁵

Temporary discontinuation of beta-blockers before TAVI has been suggested as a potential strategy to reduce the rate of PPI. The findings from observational studies investigating the effects of peri-procedural discontinuation of beta-blockers on PPI rates are conflicting. One study showed that beta-blocker continuation lowers the risk of arrhythmia and PPI after TAVI, compared with beta-blocker discontinuation.¹¹² In contrast, another study showed no difference in PPI rates between patients undergoing TAVI with or without beta-blockers.¹¹³

Early detection of conduction disturbances and risk stratification

Patient stratification according to PPI risk helps identify patients suitable for early discharge following TAVI. In a prospective

analysis of 2110 patients without prior pacemaker undergoing TAVI, the 30-day PPI rate was low (1.8%) when the pre-TAVI ECG was normal and was 10% in patients with pre-existing conduction disturbances (excluding RBBB).¹¹⁴ In that study, 78% of conduction disturbances occurred within 48 h post-TAVI, and among patients with a stable ECG 48 h post-TAVI, none developed third-degree heart block by 30 days.¹¹⁴ A retrospective analysis ($n = 1139$) detected third-degree heart block within the first week post-procedure in only 0.8% of patients with $QRS < 120$ ms post-TAVI.¹¹⁵ Similarly, a retrospective analysis of 1064 patients without a pre-procedure pacemaker who underwent TAVI found that patients in sinus rhythm with $QRS < 120$ ms and $PR < 200$ ms post-TAVI did not develop high-degree AV block by 30 days.¹¹⁶ In this regard, patients with no intra-procedural conduction disturbance, $QRS < 120$ ms and $PR < 200$ ms post-TAVI, may be safely considered for same-day discharge.

A $QRS > 120$ ms after TAVI has been associated with increased risk of delayed high-degree AV block by 30 days (8% risk in patients with pre-existing LBBB and 13% in patients with new-onset LBBB).¹¹⁶ Most of the delayed-onset third-degree heart blocks occurred within 48 h post-TAVI.¹¹⁶ In a prospective analysis of 459 TAVI patients without prior pacemaker, who were surveyed with continuous ambulatory ECG monitoring for 2 weeks post-TAVI, there was a 6.6% incidence of high-degree AV block in patients without RBBB who had new-onset conduction disturbances (LBBB, first-degree AV block, or widening of QRS and/or $PR \geq 20$ ms), whereas among patients with unchanged conduction disturbances post- compared with pre-TAVI (excluding RBBB), the incidence was 6%.¹¹⁷ Patients with unchanged pre-existing RBBB exhibited a higher (13%) rate of PPI at 2 weeks.¹¹⁷ Importantly, patients

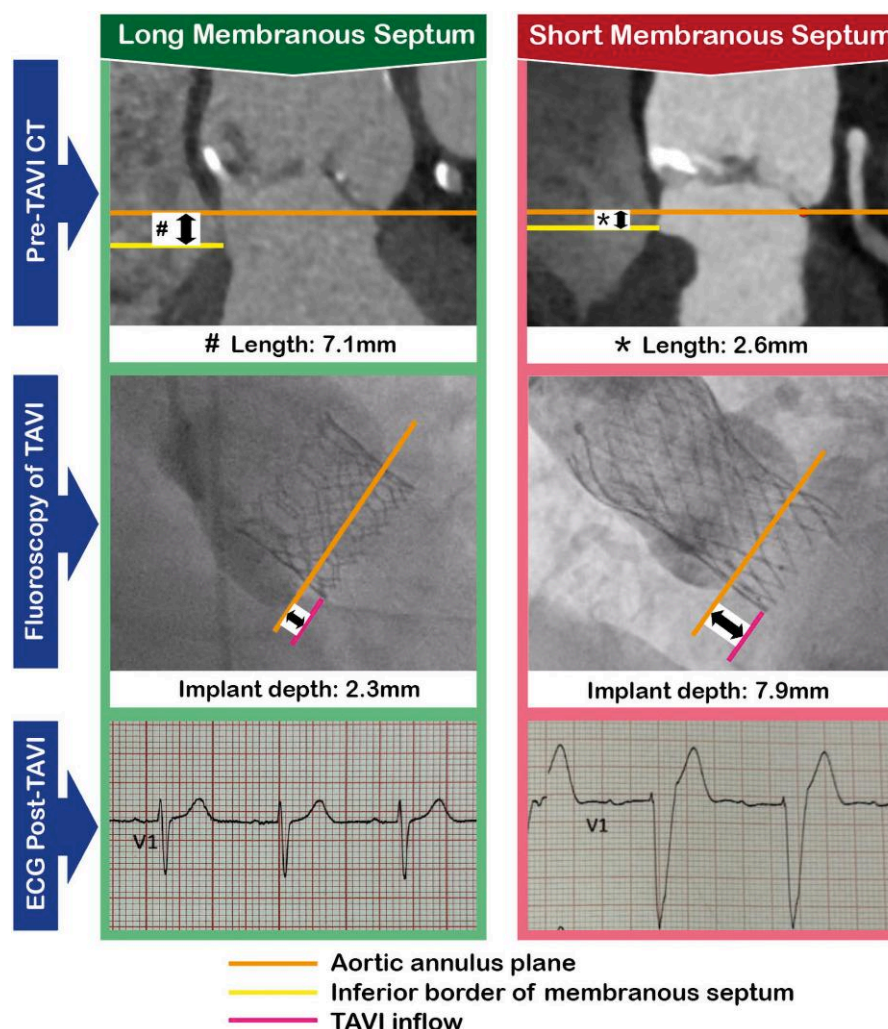


Figure 5 Case examples showing membranous interventricular septum length measured in the stretched vessel view on computed tomography, transcatheter aortic valve implantation implant depth, and the electrocardiograms performed post-transcatheter aortic valve implantation. The membranous septum is the distance from the aortic annulus plane to the most superior part of the muscular interventricular septum. The membranous septum measurements were performed using the 3Mensio computed tomography software (Pie Medical Imaging, Maastricht, Netherlands). The beginning and end of the thinnest part of the interventricular septum were identified on the annular plane image by rotating around the aortic centre line between the non-coronary cusp and right coronary cusp adjacent to the septal tricuspid valve leaflet and by visualizing the thinnest portion of the interventricular septum in the corresponding sagittal plane (stretched vessel view). The panel on the left shows a long membranous septum length, high implantation of a 26 mm SAPIEN-3 Ultra RESILIA valve, and sinus rhythm with normal QRS and PR intervals post-transcatheter aortic valve implantation. The panel on the right shows a short membranous septum length, deep implantation of a 26 mm Evolut FX+ valve, new-onset left bundle branch block and first-degree heart block. ECG, electrocardiogram; TAVI, transcatheter aortic valve implantation

with post-TAVI widening of both QRS and PR intervals (mean increases of ≥ 22 ms and ≥ 40 ms, respectively) exhibited an increased risk of delayed PPI, in a retrospective analysis that excluded patients who had PPI < 48 h post-TAVI.¹¹⁸

A prospective study stratified TAVI patients ($n = 467$) according to QRS and PR duration, to predict the risk of high-degree conduction disorders (defined as type two second-degree AV block, third-degree AV block, or atrial fibrillation with slow ventricular response) within 30 days post-procedure.¹¹⁹ None of the patients with QRS duration < 120 ms and PR interval < 200 ms post-TAVI had high-degree conduction disorders.

The risk of high-degree conduction disorders was 4.6% for patients with QRS between 120 and < 150 ms and PR between 200 and < 240 ms (excluding RBBB), and the risk was higher (13.9%) for patients with QRS ≥ 150 ms and PR ≥ 240 ms (excluding RBBB). Patients with RBBB had the highest risk of late high-degree conduction disorders (21.7%).¹¹⁹ For patients in atrial fibrillation (excluding RBBB), the risk of high-degree conduction disorders was 6.1% for those with QRS < 140 ms and 10.0% for those with QRS ≥ 140 ms. These findings suggest that decision-making about the duration of telemetry can be guided by PR and QRS durations on the 12-lead ECG post-TAVI.

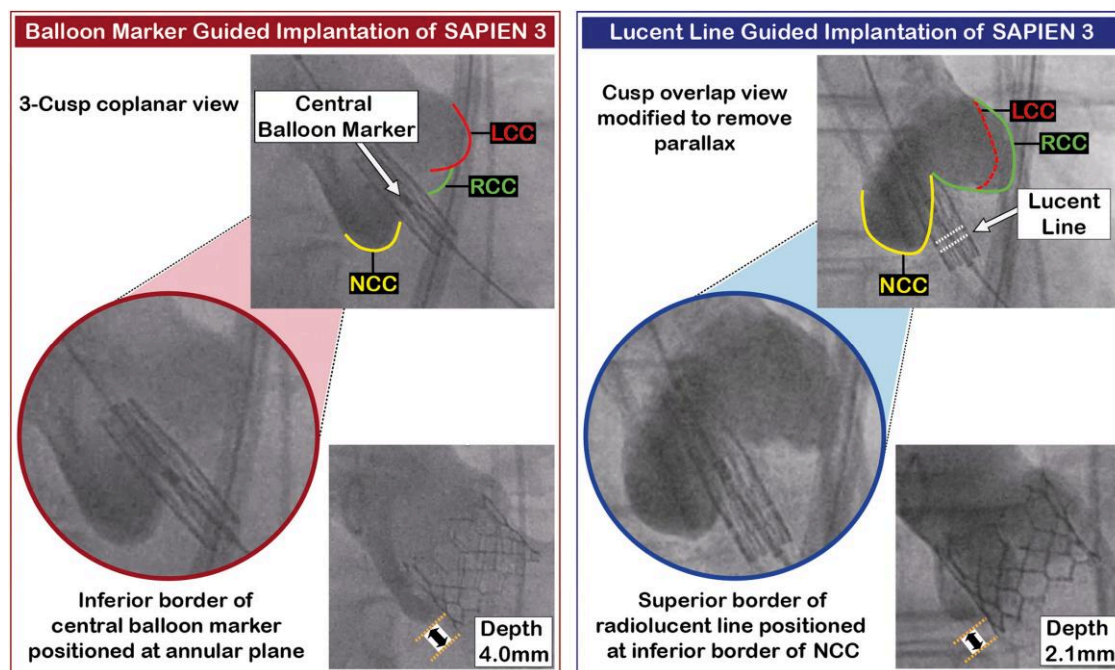


Figure 6 Case examples showing the radiolucent line implantation technique and the balloon marker guided implantation technique for deployment of SAPIEN-3 valves. Radiolucent line implantation technique: A modified two-cusp view (typically right anterior oblique caudal projection) is achieved by squaring the valve frame to remove parallax, which optimizes visualization of the radiolucent line (located at the superior aspect of the lowest set of stent struts of the crimped valve). The valve is positioned by aligning the radiolucent line with the base of the non-coronary cusp. In this example, the valve implantation depth measured on fluoroscopy was 2.1 mm. Balloon marker technique: To deploy the transcatheter heart valve, a three-cusp (coplanar) view (typically left anterior oblique caudal projection) is used, where all three cusps are seen in one plane. The valve prosthesis is positioned such that the central balloon marker is on (or slightly above) the annular plane. The valve implant depth was 4.0 mm in this example. In both cases, a pigtail catheter is placed in the non-coronary cusp to obtain an aortogram at the start of rapid pacing. LCC, left coronary cusp; RCC, right coronary cusp

New-onset RBBB after TAVI is a particularly high-risk finding.^{13,43,44} In a case series of 15 patients with post-TAVI new-onset RBBB, 67% underwent PPI after developing high-degree AV block.⁴⁴ Another analysis of 41 patients with new-onset RBBB after TAVI demonstrated a PPI rate of 46% at 1 year.¹³ Among the patients with new-onset RBBB, a PR > 228 ms and an increase in PR > 24 ms had very high risk (about 90%) of progression to high-degree AV block, and almost all episodes of high-degree AV block occurred within 1 week post-TAVI.¹³ Therefore, PPI pre-discharge may be warranted in such patients.

An alternative strategy to facilitate early discharge in TAVI patients with new-onset QRS > 120 ms and/or PR > 200 ms is continuous ambulatory ECG monitoring after discharge,¹¹⁷ since the highest risk of AV block is in the first month.¹²⁰ This strategy had a 5.7% risk of PPI for high-degree AV block (detected at a median of 5 days post-TAVI).¹¹⁷ However, there is limited data available to recommend the duration or type (ambulatory monitoring with alarm systems, patches, etc.) of ambulatory or invasive (implantable loop recorder) rhythm monitoring that would be sufficient.

Another strategy is to perform an electrophysiology study (EPS) in patients with new-onset persistent LBBB after TAVI, as a His-ventricle (HV) interval ≥ 70 ms has been

reported to independently predict progression to high-degree AV block.¹²¹ However, this association was not reproduced in other studies and the exact timing of the EPS and the optimal cut-off value of the HV interval need to be validated.^{122,123} In patients with prophylactic PPI post-TAVI for LBBB (new-onset or existing), a HV interval ≥ 70 ms was associated with only 6.5% ventricular pacing burden at median 21-month follow-up.¹²⁴ Of note, PR prolongation is often associated with a long HV interval¹²⁵; hence, the added value of an EPS when the PR is normal is controversial. A randomized trial is ongoing to investigate an EPS-guided approach to PPI post-TAVI in patients with new-onset LBBB (NCT03303612), and an ongoing observational study aims to clarify the reversibility of early conduction disturbances post-TAVI detected by EPS at 3–7 days, by performing a second EPS at 30–45 days (NCT06481137) (Tables 1 and 2).

Rapid right atrial pacing (up to 120 b.p.m.) at the end of the TAVR procedure, using the TPW, may be an easy and cost-effective strategy to further investigate and stratify the risk of PPI.¹²⁶ An observational study reported that absence of pacing-induced Wenckebach AV block during rapid atrial pacing was associated with a low likelihood of subsequent PPI.¹²⁷ However, in another study, no significant association was reported.¹²⁸

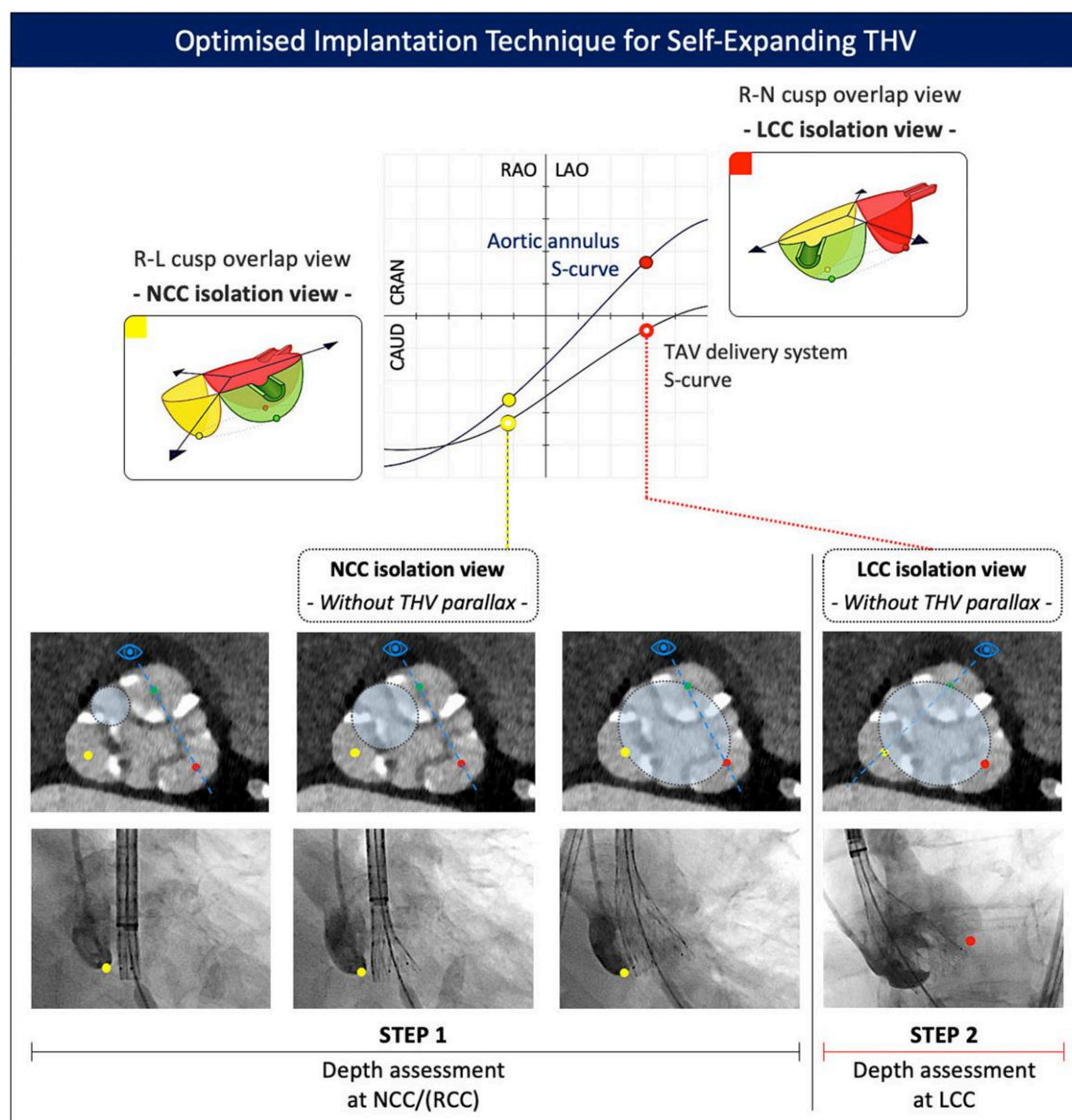


Figure 7 Case example showing cusp isolation to optimize implantation of an Evolut self-expanding valve. The right coronary cusp and left coronary cusp are overlapped (usually in a right anterior oblique caudal view) to isolate the non-coronary cusp, and parallax is eliminated from the transcatheter heart valve. This allows accurate depth assessment of the inflow of the Evolut valve frame and the nadir of the non-coronary cusp (target 1–3 mm implant depth). The Evolut valve marker dot is located 3 mm from the Evolut inflow. After the Evolut transcatheter heart valve is 80% deployed, with full annular contact, the fluoroscopic angle is moved to a left anterior oblique view to isolate the left coronary cusp (the right coronary cusp and non-coronary cusp are overlapped). Parallax is removed from the transcatheter heart valve, such that the three-valve marker dots are in line. This allows accurate depth assessment at the level of the left coronary cusp, aiming for the marker dot at the nadir of the left coronary cusp. Of note, a pigtail catheter is placed in the non-coronary cusp to obtain an aortogram when assessing the transcatheter heart valve depth. CAU, caudal; CRA, cranial; LAO, left anterior oblique; LCC, left coronary cusp; NCC, non-coronary cusp; RAO, right anterior oblique; RCC, right coronary cusp; THV, transcatheter heart valve. Reproduced from Khokhar *et al.*¹⁰⁷

Prophylactic pre-procedure permanent pacemaker implantation for right bundle branch block

In patients with pre-existing RBBB, the reported incidence of high-degree AV block during hospitalization for TAVI is 55%.⁴⁶ Of these, 77%–85% occur intra-procedurally,^{46,129} and 98%

occur within the first 3 days post-TAVI.⁴⁶ About 50%–60% of TAVI patients with pre-existing RBBB do not require PPI during the hospitalization period or during 30-day follow-up^{46,130}; however, of these patients, up to 25% may require PPI at 4 years.⁴⁶ Indeed, several studies have shown that pre-existing RBBB is associated with increased risk of delayed high-degree AV block post-TAVI.^{46,116,117,119} Analysis of pacemaker data demonstrated that the majority of high-degree conduction

Table 1 Ongoing randomized trials evaluating strategies for the prevention, detection, and management of conduction disturbances after transcatheter aortic valve implantation

Study name and estimated enrolment	Comparison	Key inclusion criteria	Key exclusion criteria	Primary endpoint
Prevention				
BETA-TAVI (NCT05721170) n = 347	Withdrawal of beta-blockers ≥ 72 h pre-TAVI and ≥ 7 days post-TAVI vs continuation of beta-blockers without interruption	Undergoing TAVI for severe AS	- PPI prior to TAVI - Pre-existing bradycardia (< 50 b.p.m.) 2nd and 3rd degree AV block	PPI at 7 days
B-TAVR (NCT06472934) n = 498	Withdrawal of beta-blockers 72 h pre-TAVI and for 72 h post-TAVI vs continuation of beta-blockers without interruption	- Undergoing elective TAVI - Indication for beta-blockers with a prior treatment duration of ≥ 1 month	- Emergency or urgent TAVI - PPI prior to TAVI - Prior intolerance of beta-blockers	Co-primary endpoints at 30 days - All-cause mortality - Heart failure hospitalization - Stroke - Arrhythmia requiring treatment
EAGLE-TAVR (NCT06762145) n = 200	IV methylprednisolone for 3 days starting from the day of TAVI vs placebo	Undergoing transfemoral TAVI	- PPI prior to TAVI - Contraindications/allergy to steroids - Chronic steroid use - Acute myocardial infarction ≤ 1 month - Gastrointestinal bleeding	Persistent LBBB at 30 days
TAVI-PACE (NCT05889208) n = 320	Prophylactic pre-TAVI PPI vs PPI after TAVI if AV block occurs	- Undergoing TAVI for severe AS - Pre-existing RBBB (QRS > 120 ms) - LVEF $\geq 40\%$	- PPI prior to TAVI - Other indication for PPI	Composite of all-cause mortality, unplanned PPI, and new-onset heart failure at 1 year
Detection				
COME-TAVI (NCT03303612) n = 250	EPS-guided approach, whereby patients with a His-ventricle interval ≥ 65 ms receive PPI vs clinical monitoring strategy, whereby patients undergo ≥ 72 h of telemetry and receive transcutaneous monitoring prior to hospital discharge for 30 days	Persistent new-onset LBBB after TAVI	- PPI prior to TAVI - Pre-existing RBBB	Composite of death, cardiovascular hospitalization, or syncope at 12 months
Pacing method				
RAFT-TAVR PACE (NCT06857201) n = 430	LBBAP vs right ventricular pacing	Persistent high-degree AV block ≤ 30 days after TAVI	- PPI prior to TAVI >Mild paravalvular leak after TAVI	Composite of cardiovascular mortality and heart failure
				<i>Continued</i>

Table 1 Continued

Study name and estimated enrolment	Comparison	Key inclusion criteria	Key exclusion criteria	Primary endpoint
Left Bundle BRAVE (NCT05541679) n = 46	LBBAP followed by right ventricular septal pacing vs right ventricular septal pacing followed by left bundle branch area pacing	- Undergone TAVI in the last 4 weeks - Symptomatic bradycardia, or symptomatic persistent AV block including first-degree and Mobitz I, or Mobitz II 3rd degree AV block	- Mechanical tricuspid valve - Renal failure (on dialysis or eGFR < 15 mL/min) - Acute coronary syndrome < 3 months - Pre-existing implantable cardiac device/PPI >Mild paravalvular leak after TAVI - LVEF < 45% - Indication for biventricular pacing - Mechanical heart valve	hospitalizations at 12 months
PACE-4 TAVI (NCT05966675) n = 500	LBBAP or His bundle pacing vs right ventricular pacing (if LVEF ≥ 40%) or biventricular pacing (if LVEF < 40%)	Undergone TAVI in the last 30 days	Pre-existing PPI	Heart failure hospitalization and all cause death at 12 months
PLANET (NCT05024279) n = 30	LBBAP vs right ventricular pacing	- Undergone TAVI for severe aortic stenosis - LVEF ≥ 50% - Guideline-based indication for pacing due to type III or II AV block, or symptomatic bradycardic AF with an expected ventricular pacing burden > 20%	- Pre-existing implanted cardiac device - LVEF < 50% - Expected pacing burden < 20%	QRS duration at 3 months
Conventional vs left bundle branch in pacing in TAVI (NCT05895097) n = 50	LBBAP vs right ventricular pacing (if LVEF ≥ 40%) or biventricular pacing (if LVEF < 40%)	- Indication for PPI after TAVI (during same hospitalization) - LVEF < 50%	- Pre-existing PPI - Ischaemic cardiomyopathy with interventricular septal fibrosis - Renal failure with eGFR < 30 mL/min	LVEF at 12 months

AF, atrial fibrillation; AS, aortic stenosis; AV, atrioventricular; BETA-TAVI, impact of beta blockers on TAVI; B-TAVR, impact of beta-blocker administration on outcome among patients undergoing transcatheter aortic valve replacement; COME-TAVI, comparison of a clinical monitoring strategy vs electrophysiology-guided algorithmic approach in patients with a new left bundle branch block after TAVI; EAGLE-TAVR, early effect of glucocorticoid on incidence of persistent left bundle branch block post-TAVI; eGFR, estimated glomerular filtration rate; EPS, electrophysiology study; GLUCO-TAVR, glucocorticoid treatment in transaortic valve replacement to reduce the incidence of conduction disturbances; Left Bundle BRAVE, comparison of left bundle branch area vs right ventricular septal pacing in patients with high-degree conduction disease after transcatheter aortic valve replacement; LBBB, left bundle branch block; LBBAP, left bundle branch area pacing; LVEF, left ventricular ejection fraction; PACE-4-TAVI, physiologic cardiac pacing to prevent left ventricular dysfunction post TAVI; PLANET, pacing of the left bundle branch area necessitated after TAVI; PPI, permanent pacemaker implantation; RAFT-TAVR PACE, randomized assessment of LBBAP vs right ventricular pacing in post-TAVI patients with persistent high-degree AV block or complete heart block requiring PPI; RBBB, right bundle branch block; TAVI, transcatheter aortic valve implantation; TAVI-PACE, prophylactic permanent pacemaker in patients with RBBB undergoing TAVI.

disturbances in patients with pre-existing RBBB did not recover at 1 month post-TAVI.⁴⁶ Notably, patients with pre-existing RBBB and persistent intra-procedural high-degree AV block had a higher ventricular pacing burden at 1 month compared with those with transient intra-procedural or post-procedural high-degree AV block (99% vs 72%).⁴⁶ Importantly, patients with pre-existing RBBB, who are discharged without PPI after TAVI, have an increased risk of early¹³¹ and 2-year¹³⁰ cardiovascular mortality compared with patients without pre-existing RBBB. Extended telemetry for a minimum of 48 h and a TPW for at least 24 h post-TAVI have been proposed for patients with pre-existing RBBB.^{132,133} However, this approach prolongs the length of hospital stay, increases the risk of TPW-related complications, and does not mitigate against the development of delayed high-degree AV block.

Some centres have adopted prophylactic PPI pre-TAVI in patients with pre-existing RBBB^{134,135} (Figure 8). In a retrospective analysis, from the UK, 62% of TAVI patients with pre-existing RBBB had prophylactic PPI at the discretion of the responsible clinician or Heart Team, resulting in shorter duration of hospitalization.¹³⁴ Among the patients with pre-existing RBBB who underwent TAVI without prophylactic PPI, 67% required urgent PPI after TAVI.¹³⁴ Two-thirds of patients with prophylactic PPI had a ventricular pacing burden > 10% in the first year post-TAVI.¹³⁴ A high pacing requirement was predicted by concomitant baseline first-degree heart block and QRS > 140 ms, but not left anterior fascicular block.¹³⁴ In another study, among patients with pre-existing RBBB, PR interval > 240 ms independently predicted PPI post-TAVI.¹³⁶

In a Danish-Dutch cohort, patients with RBBB were selected for prophylactic PPI pre-TAVI if they had QRS ≥ 150 ms, PR ≥ 220 ms, or left anterior or posterior fascicular block.¹³⁵ This selective prophylactic PPI strategy resulted in shorter procedure time, shorter length of stay, and similar mortality at 2 years when compared with historic controls without prophylactic PPI. Moreover, the prophylactic PPI strategy for pre-existing RBBB prevented emergency readmissions for PPI due to complete heart block, which occurred in 18% of patients with pre-existing RBBB who were discharged without PPI post-TAVI.¹³⁵ The ongoing TAVI-PACE (prophylactic permanent pacemaker in patients with RBBB undergoing TAVI) trial (NCT05889208) aims to clarify the efficacy of this strategy (Table 1). TAVI-PACE is randomizing patients with pre-existing RBBB to prophylactic pre-TAVI PPI, or PPI after TAVI if AV block occurs. The composite primary endpoint is all-cause mortality, unplanned PPI, and new-onset heart failure at 1 year.

Prophylactic post-procedure permanent pacemaker implantation for left bundle branch block

In patients with new-onset LBBB, the role of prophylactic post-TAVI PPI is controversial. New-onset persistent LBBB post-TAVI is associated with a 10% risk of PPI at 1 year^{120,137} and an increased risk of sudden death (particularly when the QRS is >160 ms).¹³⁸ However, new-onset LBBB is transient in at least one-third of TAVI patients.⁷⁰ Importantly, an analysis of 329 patients with PPI after TAVI found that prophylactic

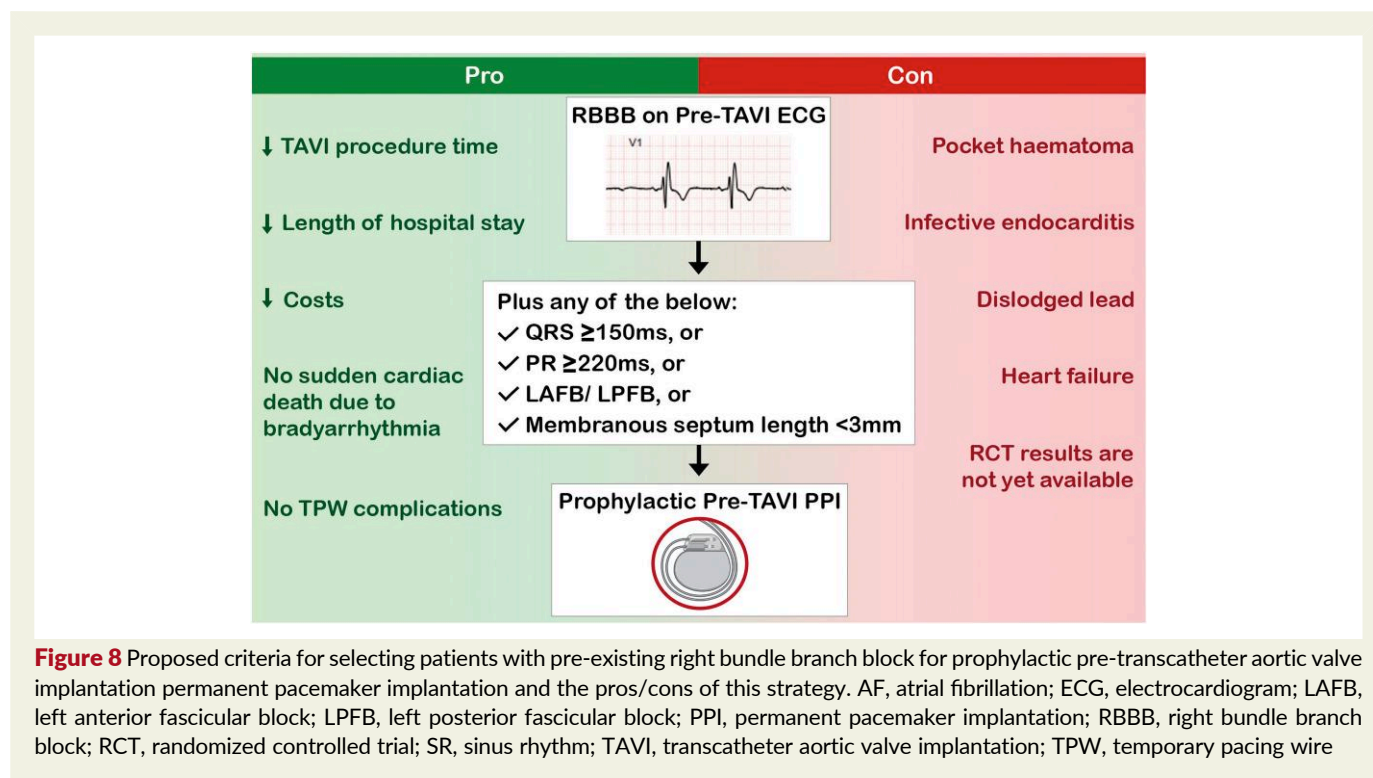
Table 2 Ongoing prospective observational studies evaluating strategies for the prevention, detection, and management of conduction disturbances after transcatheter aortic valve implantation

Study name and estimated enrolment	Study group(s)	Key inclusion criteria	Key exclusion criteria	Primary endpoint
Prevention				
MIDAS-ULM (NCT04734392) n = 300	TAVI implantation using the cusp-overlap technique, and MIDAS approach vs historical controls	Undergoing TAVI for severe AS	• PPI prior to TAVI • Valve-in-valve procedure	PPI rate at 5 days
Detection				
Rapid atrial racing after TAVI to predict PPI (NCT06189976) n = 400	Single group: rapid atrial pacing pre- and post-TAV deployment until Wenckebach AV block is observed, or until the maximum atrial pacing rate is reached	• Undergoing TAVI for severe AS • Sinus rhythm • TPW during TAVI procedure	• PPI prior to TAVI • Active infection • Persistent 2nd or 3rd degree AV block during, or pre-TAVI	PPI at 6 months
TAVR-HBE	Single group: EPS recording of His bundle electrocardiograms during TAV placement	Undergoing TAVI	PPI prior to TAVI	3rd degree heart block at 30 days
Continued				

Table 2 Continued

Study name and estimated enrolment	Study group(s)	Key inclusion criteria	Key exclusion criteria	Primary endpoint
(NCT04982406) n = 410				
TAVI-REVERSE (NCT06481137) n = 209	Positive EPS and with new-onset conduction disturbances resulting in PPI, followed by 4 weeks of ambulatory ECG monitoring, then a second EPS vs negative EPS and new-onset conduction disorders, without PPI, followed by 4 weeks of ambulatory ECG monitoring	New-onset conduction disorder that is persistent ≥ 48 h after TAVI: (i) LBBB with QRS > 150 ms and/or PR ≥ 240 ms or (ii) QRS/PR widening > 20 ms and pre-existing conduction disturbances	• PPI prior to TAVI • Pre-existing RBBB • Valve-in-valve procedure • TAVI indication is severe aortic regurgitation	For patients with a positive EPS: reversal of infra-His conduction disturbances at 30 days For patients with a negative EPS: advanced AV block at 30 days
AI-enabled cloud-based ECG diagnostic solution feasibility testing (NCT06013917) n = 75	Single group: single-lead ambulatory ECG monitor patch for 7 days of continuous recording, that transmits the data to a cloud-based platform for AI enhanced arrhythmia capture analysis, and then reports are generated for physician review	Post-TAVI patients	PPI prior to TAVI	New-onset conduction disturbances at 7 days
PACETAVI (NCT06756282) n = 90	Single group: evaluation of effective pacemaker stimulation	Patients who have TAVI and need PPI before hospital discharge for high-degree conduction disorders	PPI prior to TAVI	Pacing dependency at 1 year
LOCATE (NCT06055751) n = 50	Single group: long-term ambulatory monitoring with an implantable loop recorder for arrhythmia detection in patients who develop new-onset conduction disturbances post-TAVI that do not require urgent PPI	• Undergoing TAVI for severe AS • New LBBB (QRS > 120 ms), or new PR > 200 ms, or worsening of pre-existing 1st degree AV block by > 30 ms, or new RBBB (QRS > 120 ms), or intra-procedural 2:1 or higher-grade AV block with spontaneous improvement in 24 h	• Urgent PPI post-TAVI • Pre-existing implantable cardiac device • Pregnant patient	Incidence of bradyarrhythmias or late-onset heart block requiring PPI at 12 months New-onset AF at 12 months

AF, atrial fibrillation; AI, artificial intelligence; AS, aortic stenosis; AV, atrioventricular; ECG, electrocardiogram; EPS, electrophysiology study; LBBB, left bundle branch block; LOCATE, long-term evaluation of cardiac arrhythmia after TAVI; MIDAS, minimizing depth according to the membranous septum; MIDAS-ULM, reducing pacemaker implantation after transcatheter aortic valve replacement with modified implantation technique; PACETAVI, pacing dependence at 1-year follow-up in patients implanted with a pacemaker after TAVR; PPI, permanent pacemaker implantation; PROMOTE, prospective validation of a pre-specified algorithm for the management of conduction disturbances following transcatheter aortic valve replacement; RBBB, right bundle branch block; TAV, transcatheter aortic valve; TAVI, transcatheter aortic valve implantation; TAVI-REVERSE, investigation of the reversibility of cardiac conduction disturbances following percutaneous aortic valve implantation in patients with severe aortic stenosis; TAVR-HBE, transcatheter aortic valve replacement effects on cardiac conduction system; TPW, temporary pacing wire.



post-TAVI PPI in patients with new-onset LBBB and high-risk ECG features (mainly QRS > 150 ms or PR > 240 ms at 48 h) was associated with a very low pacing burden at 30 days compared with non-prophylactic PPI (2% vs 73%) and similar 30-day clinical outcomes.¹²² These findings do not support the systematic use of prophylactic post-TAVI PPI in patients with LBBB.

Framework for managing conduction disorders after transcatheter aortic valve implantation

The management of conduction disorders after TAVI varies widely.¹³⁹ One-third of TAVI centres do not have standardized management protocols for TAVI-related conduction disturbances.¹³⁹ It is paramount that TAVI centres establish clear protocols for managing conduction disturbances. A framework is provided (Figures 9 and 10) that builds upon a scientific consensus published in 2019¹³² and updated in 2024.¹⁴⁰

Based on registry data,¹⁴¹ we advocate PPI < 24 h after TAVI for intra-procedural persistent high-degree AV block^{118,142} rather than assess for conduction recovery for 24 h with TPW in place. This approach mitigates potential TPW related complications in patients with a low likelihood of recovery in conduction.¹⁴¹ Of note, a recent multicentre registry showed that for TAVI patients with intra-procedural persistent high-degree AV block, same-day PPI was associated with comparable peri-procedural complications, shorter hospitalization, and a similar ventricular pacing burden at 1 month (96% vs 90%) compared with patients who had deferred PPI implantation.¹⁴¹ In contrast,

the European Society of Cardiology (ESC) guidelines recommend PPI for patients with high-degree AV block that persists for 24–48 h after TAVI¹⁴³ (Table 3).

In patients with new-onset LBBB, we consider a daily increase in PR and/or QRS duration ≥ 20 ms as a predictor of high-degree AV block, which warrants an increased period of time on telemetry, backed by observational data^{13,117} and in line with previously published consensus documents.^{132,133} The ESC guidelines give this approach a class IIb recommendation.¹⁴³ In patients with new-onset LBBB and stable PR and QRS intervals (especially with PR < 240 ms and QRS < 150 ms), same-day discharge may be possible and could especially benefit healthcare systems with bed shortages. This strategy is based on observational studies that showed a low likelihood of high-degree AV block in such patients,^{114,116,118,119} but this is not formally endorsed by ESC guidelines.¹⁴³

A strategy of prophylactic pre-TAVI PPI for RBBB is not recommended by the ESC guidelines.¹⁴³ However, some centres already perform prophylactic pre-TAVI PPI for pre-existing RBBB, and criteria to select patients for this strategy, based on observational data,¹³⁵ are provided in Figure 8. Updated guidance is required after the TAVI-PACE trial is completed (Table 1). For patients with RBBB post-TAVI, who have not had prophylactic pre-TAVI PPI (including RBBB without ECG change post-TAVI, or new-onset RBBB), we suggest PPI pre-discharge if PR ≥ 230 ms, QRS ≥ 150 ms, or increase in PR and/or QRS ≥ 20 ms. This approach is backed by observational studies showing an elevated likelihood of high-degree AV block in this particular patient cohort.^{13,118} A prophylactic pacing strategy can reduce the duration of hospital stay post-TAVI and thus is likely to be cost-effective in healthcare systems where the cost of a bed-day is high. Moreover, in healthcare settings with

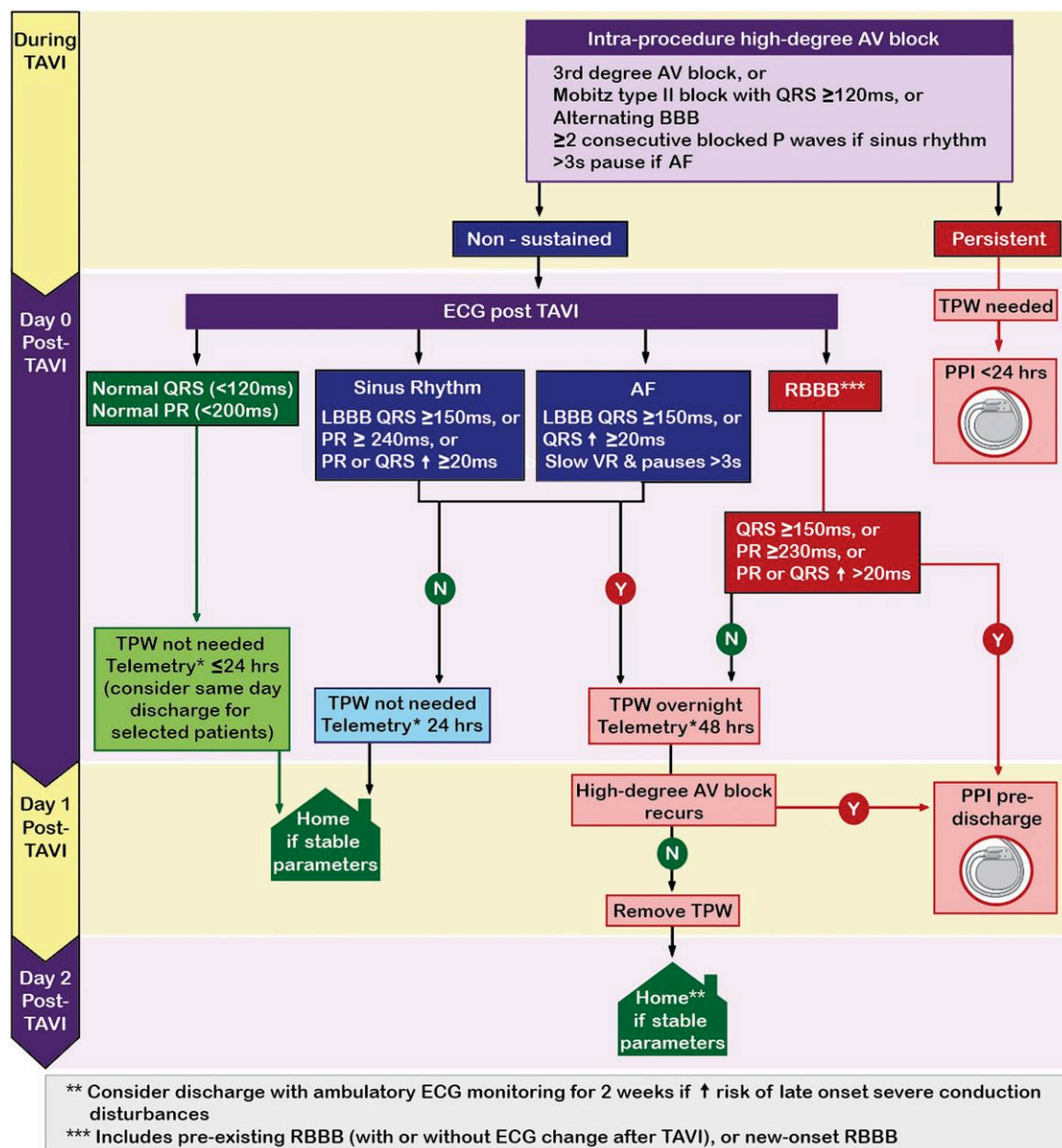


Figure 9 Proposed framework for the management of transcatheter aortic valve implantation patients with intra-procedural high-degree atrioventricular block. Persistent refers to ongoing high-degree atrioventricular block at the end of the transcatheter aortic valve implantation procedure. Non-sustained refers to high-degree atrioventricular block that resolves completely at the end of the transcatheter aortic valve implantation procedure. *Daily 12-lead electrocardiograms should be performed in addition to telemetry. **Depending on the patient's individual risk for developing delayed high-degree atrioventricular block and local resources, discharge with ambulatory electrocardiogram monitoring may be considered. AF, atrial fibrillation; AV, atrioventricular; ECG, electrocardiogram; LBBB, left bundle branch block; N, no; PPI, permanent pacemaker implantation; RBBB, right bundle branch block; TAVI, transcatheter aortic valve implantation; TPW, temporary pacing wire; VR, ventricular rate; Y, yes

bed shortages early discharge from hospital creates possibilities to increase the number of TAVI procedures performed.

For patients with LVEF $< 40\%$ who develop high-degree AV block after TAVI, cardiac resynchronization therapy (CRT) remains the standard approach in most centres. Whilst CRT in such circumstances reduces the risk of sudden cardiac death, the decision to implant a device in patients with LVEF $< 40\%$ who have RBBB and high-risk ECG features without

high-degree AV block after TAVI is less clear. Right bundle branch block is considered a predictor of worse response to CRT, and therefore, a strategy of prolonged ECG monitoring may be preferable in such patients, as high-degree AV block might not occur and LVEF may improve after TAVI, mitigating the need for CRT. Updated guidance is required, based on randomized trials, or extensive registries, so that patients with conduction disturbances post-TAVI can be optimally managed.



Figure 10 Proposed framework for the management of conduction disturbances, according to the 12-lead electrocardiogram performed after transcatheter aortic valve implantation, in patients without intra-procedural high-degree atrioventricular block. *Daily 12-lead electrocardiogram should be performed in addition to telemetry. In patients with ongoing prolongation of PR or QRS intervals ≥ 20 ms at 48 h, but without high-degree atrioventricular block, telemetry > 48 h may be considered. **Depending on the patient's individual risk for developing delayed high-degree atrioventricular block and local resources, discharge with ambulatory electrocardiogram monitoring may be considered. AF, atrial fibrillation; AV, atrioventricular; ECG, electrocardiogram; LBBB, left bundle branch block; N, no; PPI, permanent pacemaker implantation; RBBB, right bundle branch block; TAVI, transcatheter aortic valve implantation; VR, ventricular rate; Y, yes

Table 3 European Society of Cardiology guidelines¹⁴³ for pacing in transcatheter aortic valve implantation patients

ESC 2021 guidelines	Class (level of evidence)
PPI recommended if complete heart block persists for 24–48 h post-TAVI	I (B)
PPI recommended if alternating BBB post-TAVI	I (C)
PPI within 24 h post-TAVI should be considered if pre-existing RBBB and further conduction disturbance after TAVI (e.g. transient high-degree AV block, PR prolongation, or QRS axis change)	IIa (B)
Ambulatory ECG monitoring# or an EPS ^a should be considered if new LBBB with QRS > 150 ms, or PR > 240 ms, with no further prolongation during >48 h post-TAVI	IIa (C)
Ambulatory ECG monitoring# or an EPS ^a may be considered if pre-existing conduction abnormality, with further prolongation of QRS or PR > 20 ms	IIb (C)

ECG, electrocardiogram; LBBB, left bundle branch block; PPI, permanent pacemaker implantation; RBBB, right bundle branch block; TAVI, transcatheter aortic valve implantation.
^aElectrophysiology study (EPS) with a His-ventricle cut-off ≥ 70 ms may be considered for offering PPI. If an EPS is performed, a delay of ≥ 3 days is suggested for conduction abnormalities to stabilize after TAVI.
[#]Implantable or external for 7–30 days.

Pacing strategies following transcatheter aortic valve implantation

It is worth highlighting the importance of conduction system pacing [either His pacing or left bundle branch area pacing (LBBAP)] to achieve physiological pacing. The emerging LBBAP technique^{144,145} preserves physiological ventricular activation and thereby prevents pacing-induced cardiomyopathy, adverse cardiac remodelling, and the increased risk of heart failure associated with right ventricular pacing. Small studies indicate that LBBAP pacing post-TAVI results in QRS shortening when compared with right ventricular pacing^{144,145} and may reduce the risk of heart failure hospitalization, but not all-cause death at 5 years post-TAVI.¹⁴⁵ Conduction system pacing has potential to reduce long-term healthcare costs associated with heart failure hospitalizations; however, it is not widely available and large randomized trials with extended follow-up are needed to evaluate whether this technique reduces pacing-induced cardiomyopathy (Table 1). Notably, leadless pacemakers could offer advantages compared with conventional transvenous systems, such as no device pocket, lower risk of infection, implantation immediately after the TAVI (same procedure) in patients with complete heart block not deemed to recover, and theoretically less interference with the tricuspid valve apparatus, which could lower the risk of worsening tricuspid regurgitation.¹⁴⁶ New

devices can now offer single or dual-chamber leadless pacing. In some patients with significant LV dysfunction, pacing dependency, or persistent LBBB, CRT may be considered.

Future perspectives

The main focus of ongoing research in this field is preventive strategies, early detection of conduction disturbances, and optimal permanent pacing techniques (Tables 1 and 2). Innovative strategies to detect patients likely to need PPI include the CARA monitor for real-time continuous ECG analysis of conduction disturbances during TAVI.¹⁴⁷ Using this system, 83% of patients exhibited ≥ 1 intra-procedural conduction disturbance (usually during pre-dilatation or valve implantation), 11% exhibited third-degree, or Mobitz II second-degree heart block (90% after device deployment), and the PQ and QRS intervals progressively increased throughout the procedure.¹⁴⁷ Interestingly, artificial intelligence models that integrate computed tomography with clinical ECG and echocardiography data can robustly predict PPI post-TAVI (area under the receiver operating characteristic curve 92%).¹⁴⁸ However, external validation is necessary to assess the reproducibility and generalizability of such machine learning models for risk prediction in different populations. The results of ongoing randomized trials of peri-TAVI interruption of beta-blockers, methylprednisolone pre-TAVI, and LBBAP are awaited (Table 1). Lastly, innovative TAVI device technologies are needed to reduce the incidence of conduction disturbances, e.g. THVs that can be implanted in very high positions without the risk of embolization and THVs with integrated steerable delivery systems to achieve better coaxiality.

Conclusions

Conduction disturbances after TAVI remain common and constitute a significant challenge in the peri-procedural management of TAVI patients. This article presents a framework for the management of conduction disturbances after TAVI based on the available evidence. Further data are needed from randomized trials and large observational studies to optimize the prevention and management of this important clinical issue.

Supplementary data

Supplementary data are not available at European Heart Journal online.

Declarations

Disclosure of Interest

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Data Availability

No data were generated or analysed for or in support of this paper.

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Correction

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Correction to: The Safe Hearts Plan for Europe and beyond

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The following change has been made to the originally published paper. The caption to Figure 2 is corrected in the date to read: “10 December 2025” instead of: “10 December 2026”.

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